

COURSE

OVERVIEW

BASIC FALL ARREST

The Basic Fall Arrest course is designed to provide learners with the essential knowledge and practical skills required to work safely at height using fall arrest systems. The course focuses on the correct selection, inspection, use, and limitations of fall arrest equipment, as well as the principles that influence safe system setup and use.

Learners are introduced to work-at-height systems, anchor points, fall factors, minimum free space, and emergency considerations. Emphasis is placed on hazard awareness, correct equipment use, and personal responsibility to reduce the risk of injury in fall-risk environments.

BY COMPLETING THIS COURSE, LEARNERS WILL BE ABLE TO:

- Identify fall-risk environments and basic work-at-height hazards
- Select, inspect, use, and store fall arrest equipment correctly
- Explain the limitations of fall arrest systems and applicable regulations
- Identify suitable anchor points for fall arrest applications
- Use fall arrest systems safely, including double-lanyard systems
- Apply basic fall arrest principles to reduce fall distance and injury risk

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GRAVITY

VALUES



R

RELATIONSHIPS

Creating and maintaining meaningful relationships with all stakeholders of Gravity.



I

INTEGRITY

Conducting day-to-day business with integrity at all times.



S

SOLUTIONS **DRIVEN**

Being solution driven.



Q

QUALITY & **EXCELLENCE**

Striving for excellence and constant improvement.

WELCOME, LEARNER!

Working at height is inherently risky — and this includes training exercises. The best way to stay safe is by learning and practising the correct techniques. This manual will help you:

- Understand the basics of fall arrest and basic rescue
- Learn how to identify and manage height-related hazards
- Build the skills needed to work safely and confidently

Working at height means doing any job where you could fall from one level to another. This includes tasks done:

- Above the ground or floor
- Near open edges
- Near fragile surfaces
- Next to holes or gaps in the ground or floor
- Close to excavations

Note: Work at height does not include slipping or tripping hazards on the same level.

HOW TO USE THIS MANUAL?

This guide supports the Fall Arrest and Basic Rescue Course and is meant to be used with the help of a trained instructor.

At Gravity Training, all facilitators are highly trained to deliver top-quality learning experiences. The content has been designed to be clear, easy to remember, and practical for real worksites.

Before any work begins, the site must be safe.

Use this simple checklist

Follow the Rules:

Make sure you follow all client rules.

Example: If night work is not allowed, do not work at night.

01



Check Equipment

All tools and gear must be safe and working before use.

04



Rescue Plan

Have a rescue plan ready. A rescue kit and first aid kit must be on site.

02

Work in Pairs

There must be at least two workers on site.

If something goes wrong, one can help or call for help.

03



Safe Site

Check the building or structure.

Make sure it is strong and safe to work on.

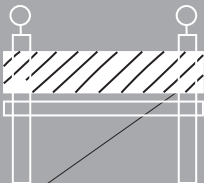
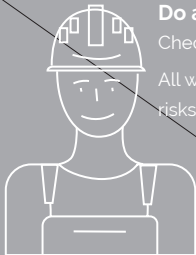
05



06 Manage Traffic

If there are cars or trucks nearby, there must be a traffic safety plan.

06



07 Do a Risk Assessment

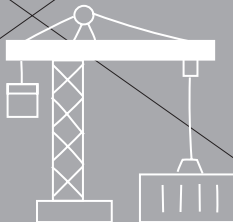
Check what dangers there are.

All workers must understand the risks before starting work.

07



08



08 Use the Right Safety Gear

Make sure the fall protection fits the job and the site.

EXCLUSION ZONES & BARRICADES

Exclusion Zone: An area where no one is allowed to walk or work.

Example: A 2-metre space around a hole in the roof.

Barricade: A physical barrier to block people from danger.

Example: A fence or tape to stop people from falling.

SITE SAFETY



Be healthy and ready before starting work.



Only do jobs you are trained and allowed to do.



Never work under the influence of alcohol or drugs.



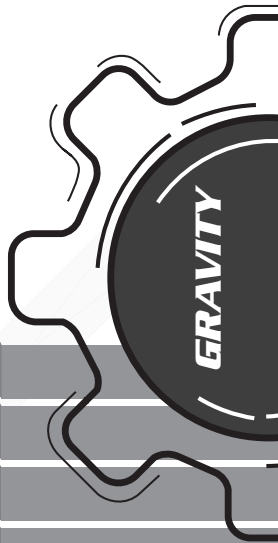
Always wear your safety gear (PPE, harness if working at height).



Use harnesses correctly and stay connected at heights.



Never work alone and always follow safety steps.



SITE SAFETY

Keep people away from areas below when working above.



Don't walk under lifted loads or stand in dangerous areas.



Only trained people can do electrical, underground, or confined space work.



Follow road rules. Wear helmets. No passengers on open trucks..



Don't drive tired or distracted.



Stop unsafe work! Everyone has the right to speak up.



A. WORK-AT-HEIGHT EQUIPMENT

1. *Personal Protective Equipment (PPE)*

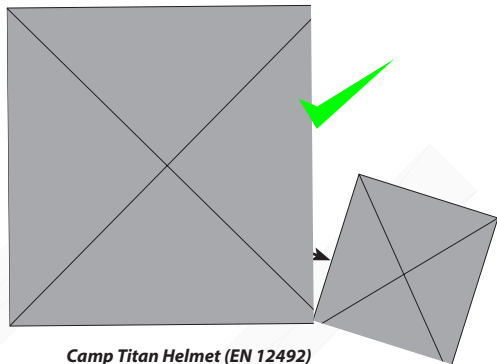
In order for a user to use PPE safely and correctly, there are four things each user must know about each piece of equipment:

- **Certification** – Only equipment with the correct European National Standards (EN) should be used. Certified equipment must be purchased through a reputable retailer such as Gravity Training. Gravity Training only makes use of ISO 9001:2015-certified equipment suppliers. All fall arrest equipment must be identified and certified to ensure quality and that equipment is safe for use and fit for purpose.
- **Identification** – The name and purpose of each piece of equipment.
- **Limitation** – How to use and not use equipment. The safe-use criteria and hazards to identify when using each piece of equipment. The maximum load that each piece of equipment is certified to carry. The load-bearing ability of equipment is given by the Original Equipment Manufacturer (OEM) as the Safe Working Load (SWL) and Minimum Breaking Strength (MBS), and should never be exceeded.
- **Inspection** – Pre-use inspection(s) should be done before the start of work, and equipment should be sent to the equipment controller for detailed inspections at least every six months. The equipment controller should be competent and appointed in writing. The use of faulty or damaged equipment may lead to serious injury or death.

Note that the SWL will differ from one manufacturer to another.

2. *Climbing Helmet*

- Used to protect a technician's head during a fall as well as from falling objects.
- Must have a chinstrap (to help keep the helmet on the worker's head during a fall or in high-wind conditions).
- Must be on the worker's head at all times.
- The chinstrap is designed to break under a force of 50 kg.
- Designed in a way that it can take top and side impact.

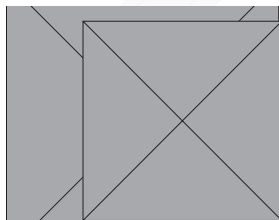


Camp Titan Helmet (EN 12492)

Construction helmet:

- Should not be used for working at height.
- Not rated for side or back impact forces.

Should an object fall on the front of the helmet, it will lift up the back of the helmet and expose the user's head to potential injury.



Construction Helmet (EN 397)

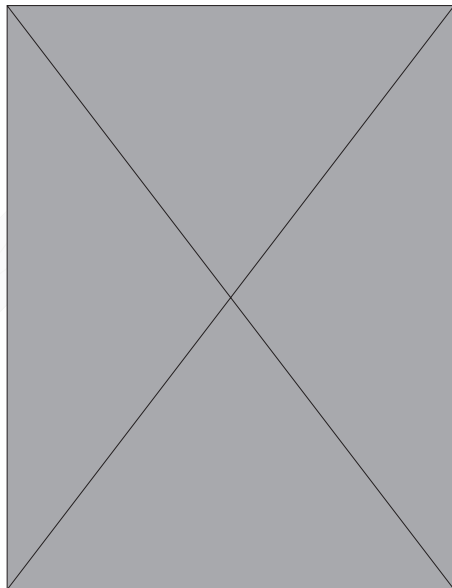
3. **Full Body Harness**

The harness allows all other devices and systems to be connected to the worker. This forms the basis of all safety systems.

- Fall arrest systems must preferably be connected to the chest D-ring.
- The two buckles on the side of the harness are for work positioning systems.
- The harness must be fitted correctly with all straps tightened properly.
- Ensure that the buckles go to the correct corresponding connectors.

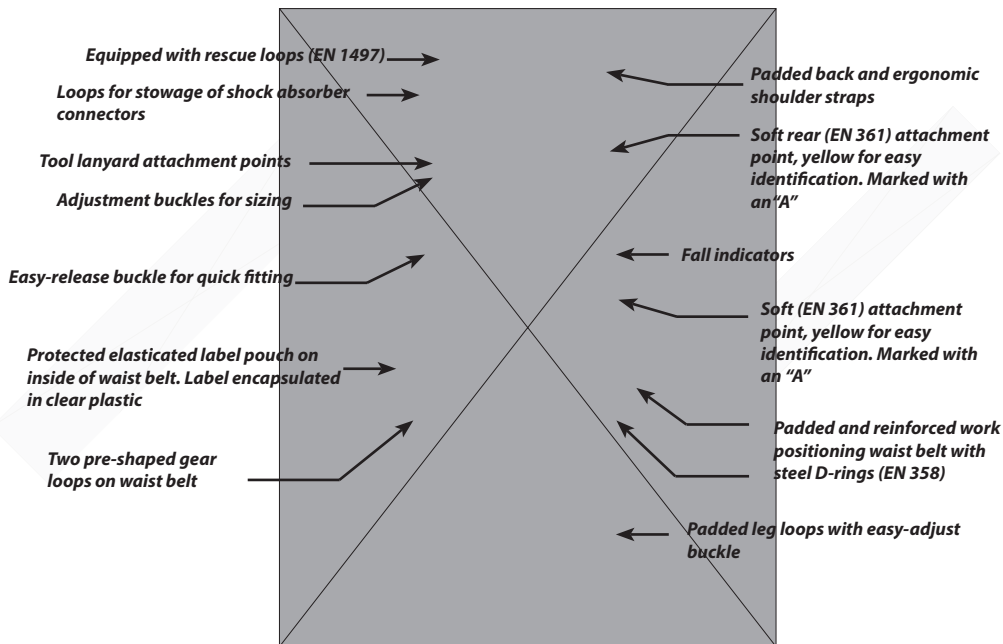
Gravity Full Body Harness:

- SWL = 140 kg
- MBS = 1 400 kg



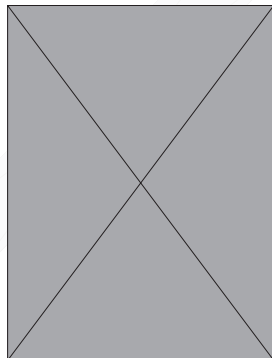
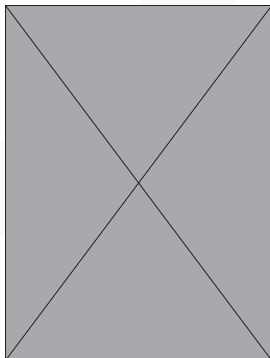
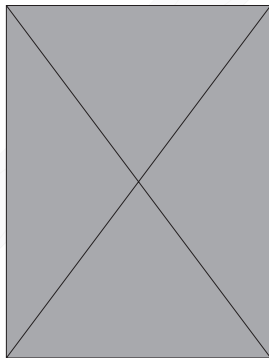
Gravity Gear FA Full Body Harness (EN 361 & 358)

H-shape fall arrest harness with speed buckle for easy fitting. Padded shoulder straps, leg loops and work positioning belt. CE and EN certified.



When donning a harness ensure that:

- the waistbelt is above your hips and is tight (but comfortable),
- the shoulder straps are fitted so that you can fit a fist between your chest and the harness,
- the leg loops are adjusted so that they do not restrict blood flow when you crouch, and
- all the buckles are connected and secure.



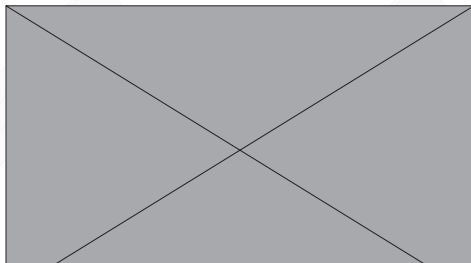
Donning and duffing of a harness

4. **Shock Absorbing Lanyard**

- Commonly known as a “shock absorber”.
- Acts as the **primary** safety device when working at height.
- Rips open (deploys) during a fall to reduce the shock load felt by the person falling.
- Must be intact during use. Once activated, the shock absorber may not be used again as it will not be able to absorb the shock load of another fall.
- You can reduce your falling distance with shorter Shock Absorbing Lanyards.

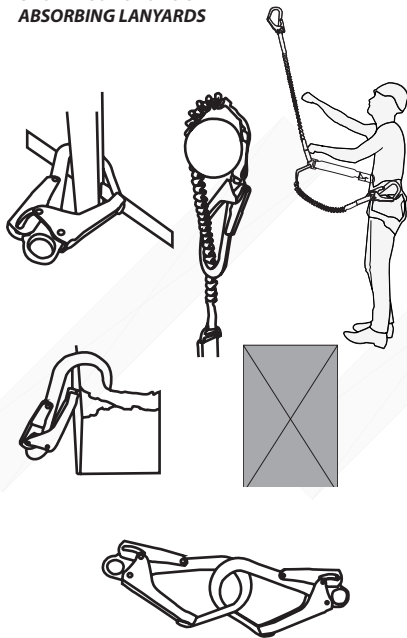
Gravity Shock Absorbing Lanyard:

- SWL = 130 kg
- MBS = 1 300 kg

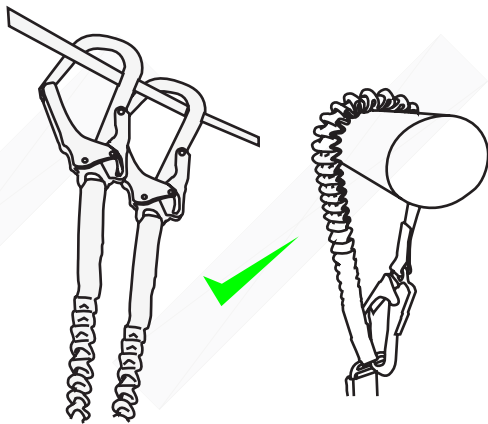


Gravity Shock Absorbing Lanyard (EN 355)

**UNSAFE USE OF SHOCK
ABSORBING LANYARDS**

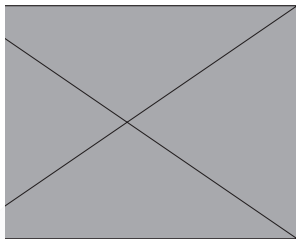


**SAFE USE OF SHOCK
ABSORbing LANYARDS**

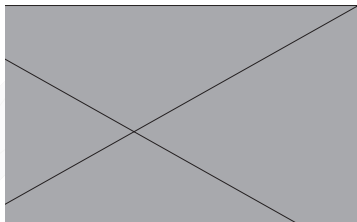


How to fit the Gravity Shock Absorbing Lanyard

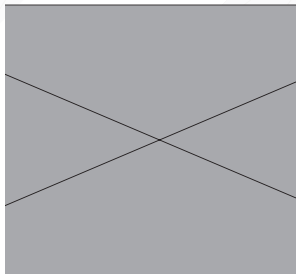
Step 1



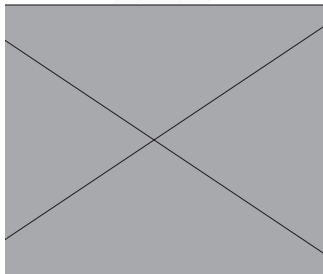
Step 2



Step 3



Step 4

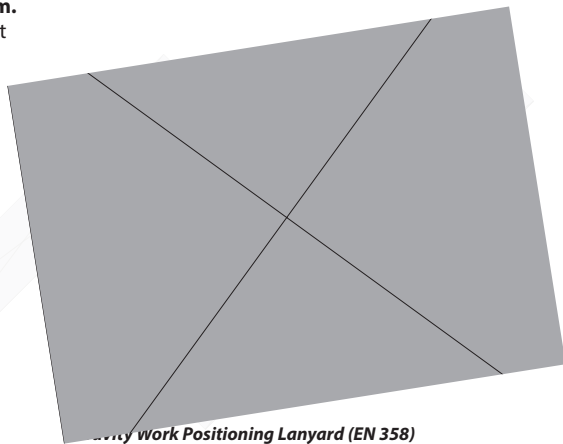


5. **Work Positioning Lanyard**

- Allows a worker to use both hands freely in order to work safely.
- **Must never be used without a fall arrest system.**
- Should be wrapped around a structure to prevent slipping.

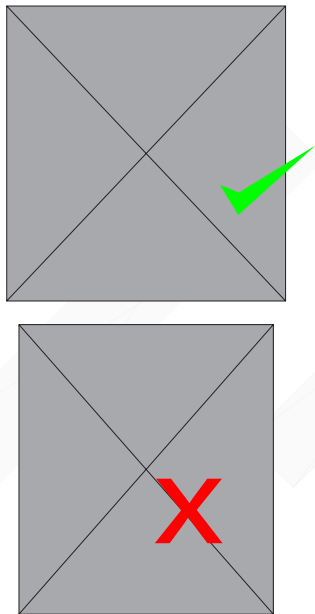
Gravity Work Positioning Lanyard:

- SWL = 150 kg
- MBS = 1 500 kg

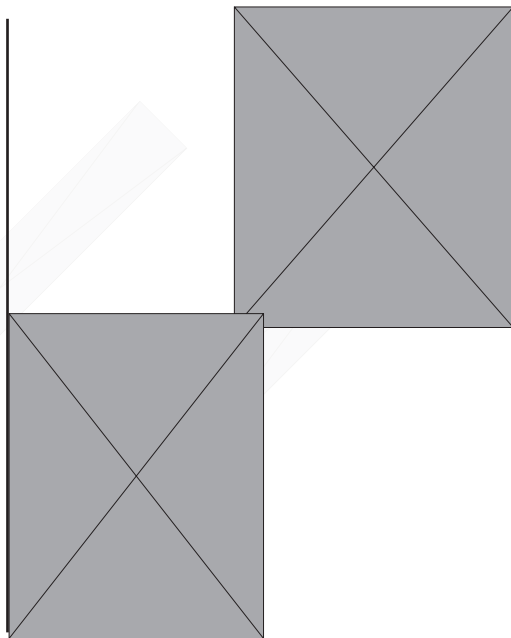


The helmet, harness, shock absorber and work positioning lanyard form the minimum PPE that must be used by a worker when at height.

How to fit the Gravity Work Positioning Lanyard



How to adjust the Gravity Work Positioning Lanyard

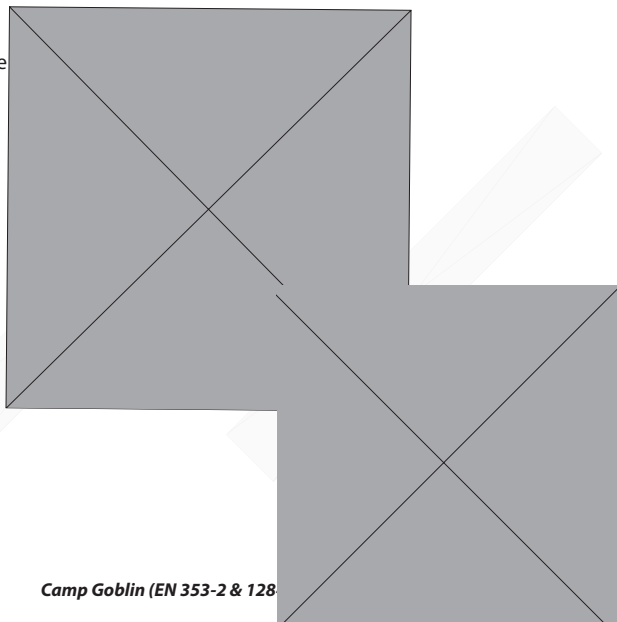


6. **Rope Grab Device**

- Used on a vertical lifeline as a fall arrest device. This will grab the rope in the event of a fall, limiting the falling distance to rope stretch only.
- Must be used in a safe Fall Factor.
- Must not be manipulated directly, as this may interfere with the working action of the device.

Camp Goblin:

- SWL = 150 kg (rescue use max 200 kg)
- MBS = 1 500 kg



Camp Goblin (EN 353-2 & 128)

7. ***Descending Device***

- A variable-friction device that controls the rate of descent.
- Used for emergency escapes, horizontal lifelines and rescues.
- Must be locked when not in use.
- Requires extra friction to be applied during rescues.

Camp Druid:

- SWL = 120 kg (rescue use max 200 kg)
- MBS = 1 200 kg

Camp Druid (E

8. *Slings*

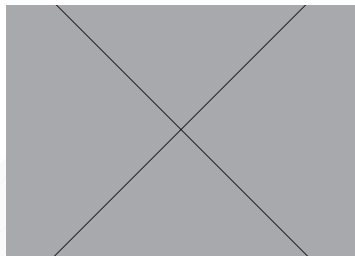
- Used to anchor systems to structures when sharp edges are not present.
- Should be wrapped around the structure to prevent slipping.
- Must be used in such a way that the stitching is not exposed to unnecessary friction.
- Available in different sizes (60 cm and 120 cm).
- Anchor slings (slings with PVC sleeves) should be used when sharp edges are a concern.

Camp Express Sling:

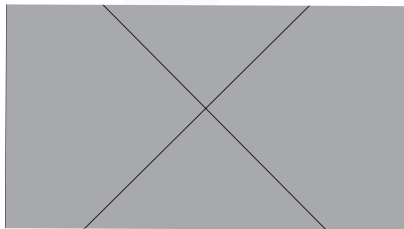
- SWL = 250 kg
- MBS = 2 500 kg

Gravity Anchor Sling:

- SWL = 220 kg
- MBS = 2 200 kg



Camp Express 60 cm Sling (EN 354, 566 & 795 B)



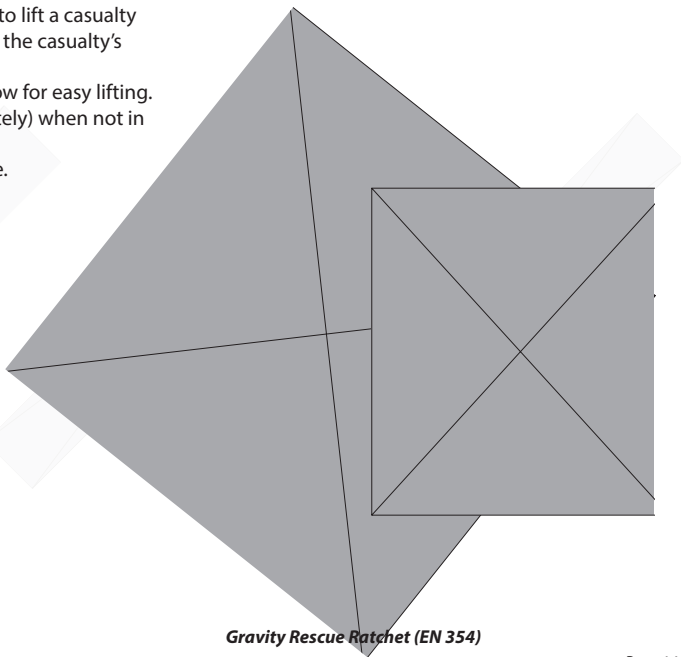
Gravity Anchor Sling (EN 354 & 795 B)

9. *Rescue Ratchet*

- Used during the ratchet rescue to lift a casualty up to allow a rescuer to remove the casualty's lanyards.
- Has a ratchet mechanism to allow for easy lifting.
- Must be locked (closed completely) when not in use.
- Should always be reset after use.

Gravity Rescue Ratchet:

- SWL = 120 kg
- MBS = 1 200 kg



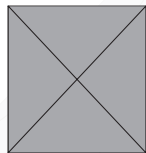
Gravity Rescue Ratchet (EN 354)

10. Connectors

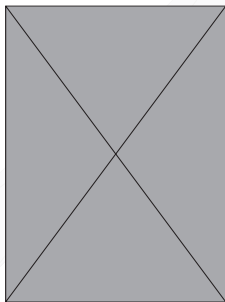
- Used to connect a range of devices and different pieces of equipment.
- There are a variety of different connectors for work at height: for example, karabiners/carabiners, maillons, snap hooks, anchoring devices, etc.

Camp Steel Oval Karabiner:

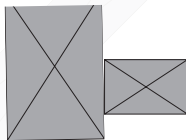
- SWL = 280 kg
- MBS = 2 800 kg



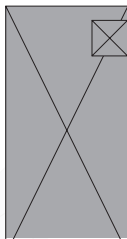
***Opposite
and
opposed***



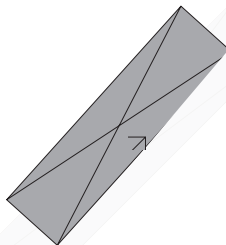
***Camp Steel Oval
Karabiner
(EN 362)***



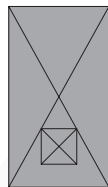
Gate facing downward



Cross-loading



Chain-linking



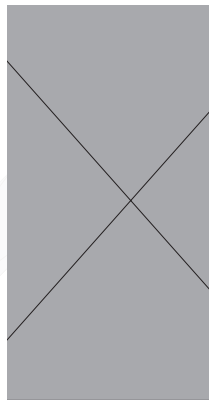
Open gate

11. **Work Rope/Semi-Static Rope**

- Used throughout work-at-height systems and a crucial part of safety at heights.
- Stretches between 5% and 10%.
- Kermantle construction consisting of two parts:
kern (core) = 80% of the strength, and
mantle (sheath) = 20% of the strength.

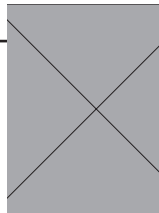
Gravity Gear Kernmantle Rope (10.5 mm):

- SWL = 290 kg
- MBS = 2 900 kg



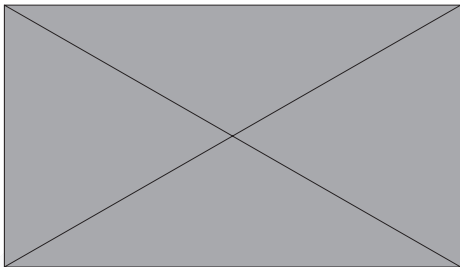
Work Rope (EN 1891)

Indicator stating date of manufacture



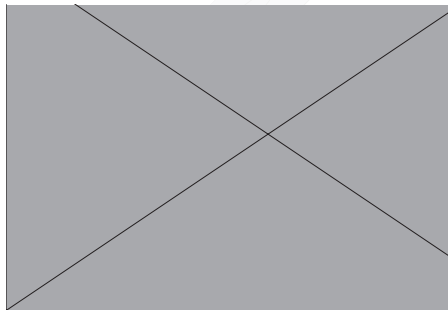
Rope care

- Do not use a rope for anything other than work at height.
- Do not run a rope over a sharp edge. Use edge protection when sharp edges are unavoidable.
- Do not stand on or walk over a rope.
- Ensure that all devices are compatible with each other.



Rope bagging

It may be preferable to bag a rope instead of coiling it. In such cases, it is recommended to deviate the rope over a structure or ladder and pull the rope into the bag. This will prevent knotting and allows the rope to be bagged faster.



12. Retractable Lanyard

Used as a self-retracting fall arrest device. Allows quick and easy movement up and down. Limits falling distance to approximately 30 cm.

- Similar functionality as a car safety belt; it allows a technician to freely move up and down and only locks when a person falls.
- Can be used for work positioning and fall arrest.
- Some models include a shock absorbing function.
- Has a 20 degree maximum range during vertical use.
- Never use above a Fall Factor 1.

Camp Cobra Retractable Lanyard:

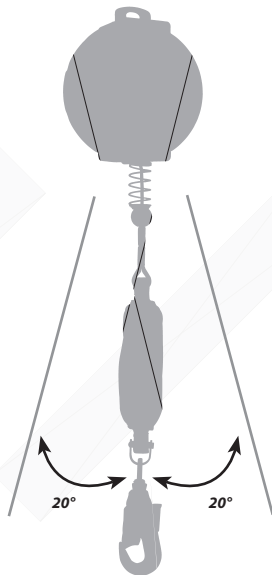
Webbing Lanyard:

- SWL = 120 kg

Cable Lanyard:

- SWL = 150 kg

Note that the SWL might differ depending on manufacturer and design.



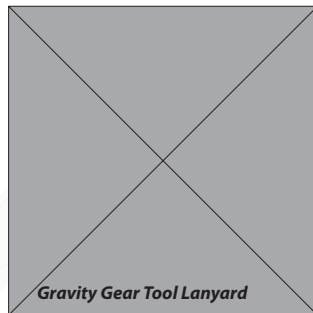
***Camp Cobra Retractable Lanyard
(EN 360)***

13. Tool Lanyard

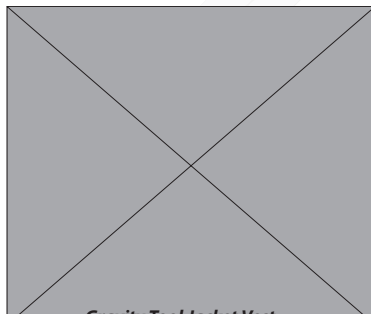
All tools must be tied off with tool lanyards or accessory cords to prevent tools from being dropped from height. Tools being dropped can cause damage to property and even injury or death to persons below.

Tools weighing 8 kg or less should be connected to the harness gear loops. Tools weighing in excess of 8 kg should be lifted using a separate 1:1 pulley system with a maximum lifting capacity of 20 kg.

Any tools or equipment weighing more than 40 kg need to be lifted by a qualified rope rigger. A qualified rope rigger may lift up to 20 kg using a 1:1 system and up to 100 kg using a 3:1 system.



Gravity Gear Tool Lanyard

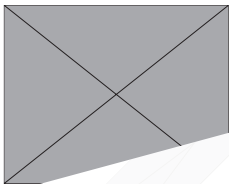


Gravity Tool Jacket Vest

14. Pre-Use Inspections

Users must inspect all the equipment to be used, before work starts. This must be done to ensure that all equipment is safe for use and fit for purpose in accordance with ISO 22846-2.

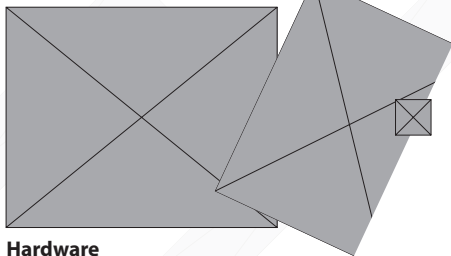
Equipment is divided into two categories, namely, hardware and software. Each of these have their own inspection criteria. Some pieces of equipment (e.g. a harness) fall into both categories and must be inspected as such.



Software

Check for:

- Date of manufacture (DOM)
- Condition of stitching
- Discolouration
- Cuts
- Burns
- Wear and tear
- Certification



Hardware

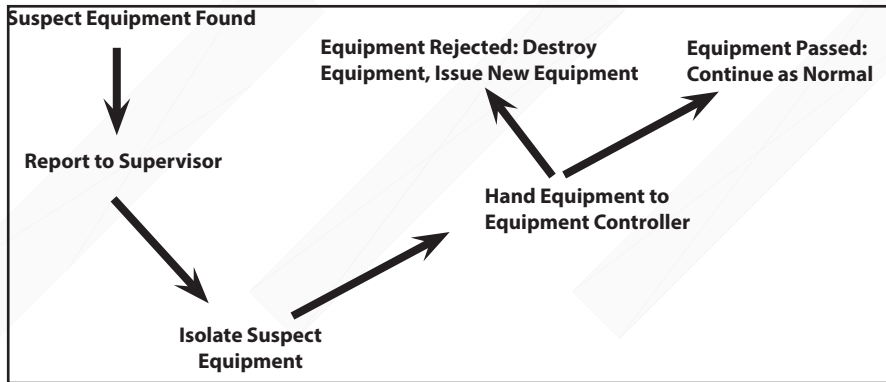
Check for:

- Rust
- Working action
- Movement
- Wear and tear
- Deformation
- Cracks

15. Reporting Suspect Equipment

Suspect equipment must be reported to the person responsible for the site (e.g. the site supervisor or safety officer).

Damaged and expired equipment must be properly marked and removed from service immediately. Rejected equipment must be destroyed at the earliest opportunity to prevent the equipment from getting back into circulation.



Repairing Damaged Equipment

Equipment may only be repaired by either the manufacturer or a company certified by the manufacturer to do so. Repairs conducted by unauthorized parties may lead to equipment failure which can cause injury or death.

No contractor may repair PPE.

16. Equipment Storage

Equipment must be stored correctly according to the manufacturer's specifications to prevent damage to equipment when not in use.

Things to consider when storing equipment in a SAFE way:

- **Secure location.** This will prevent unauthorized personnel from having access to equipment and potentially damaging equipment.
- **Away from direct sunlight.** Equipment should be stored in a cool, dry place. This will help prevent mildew in software and rust in hardware.
- **Free of chemicals.** This will reduce the chance of chemical damage to equipment.
- **Ensure that there are no infestations** which can lead to damage of software (e.g. rats and moths). Maintain a system to prevent infestations.

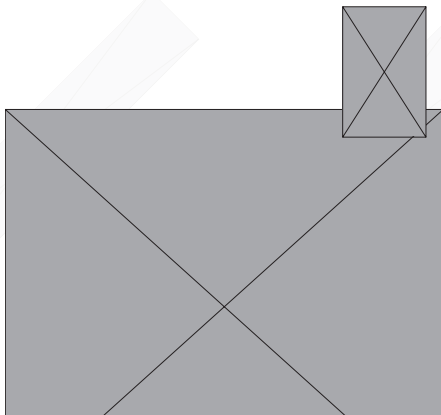
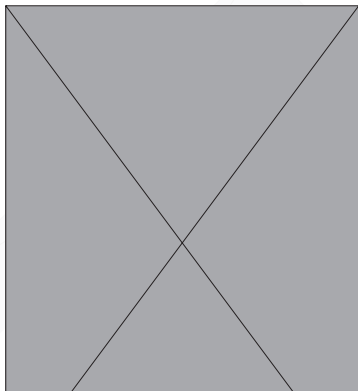
17. Housekeeping

Whilst onsite, it is important to ensure that the equipment being used is handled in such a way that it does not suffer any unnecessary damage or exposure to environmental conditions. When equipment is not in use, store it in the equipment bags or bins. Ensure that equipment is moved away from busy areas. This will help to prevent slips and trips or other accidents.

B. WORK-AT-HEIGHT SYSTEMS

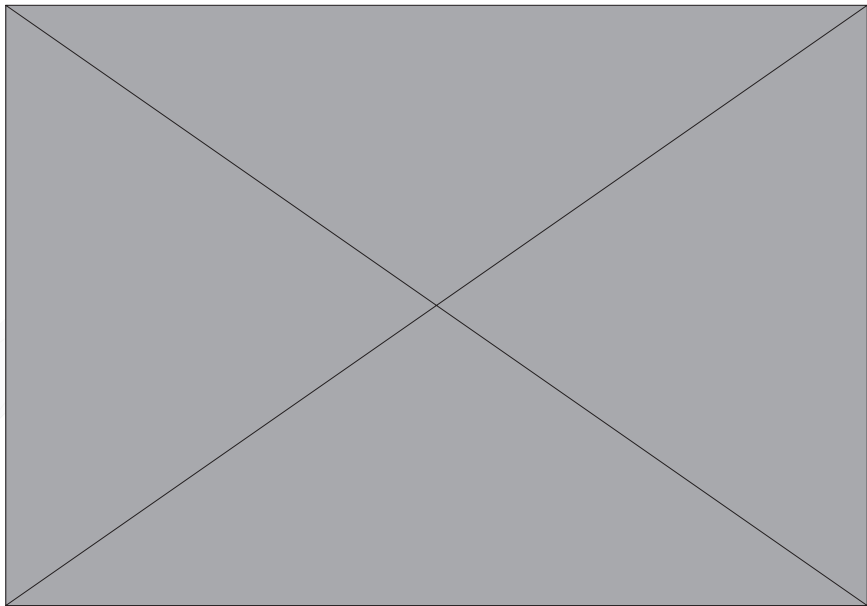
1. *Work Restraint*

A system where a worker is connected to a structure in such a way that it prevents them from reaching the edge, thus, preventing a fall.



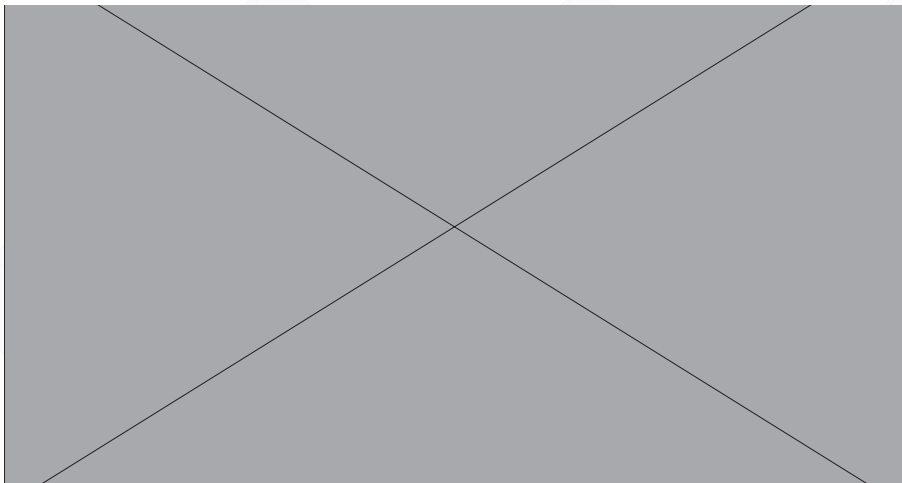
2. Rooftop Work Restraint System

One can combine a horizontal lifeline with a work restraint system to create a system ideal for rooftop work.



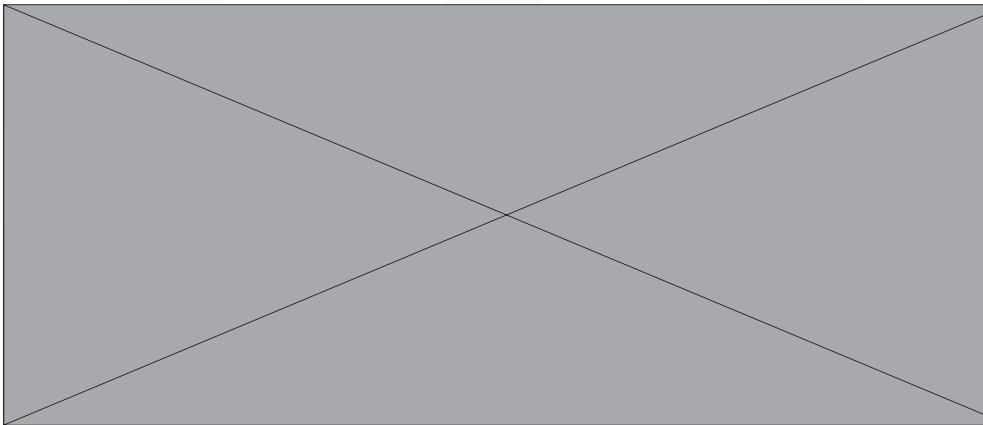
3. *Pitched Roof Work*

You can combine a temporary horizontal lifeline with a retractable lifeline to create a system ideal for pitched roof work.



4. *Using A Retractable Lanyard While Working On Containers*

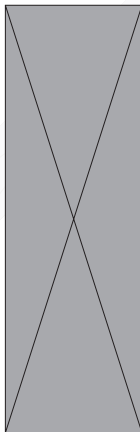
- When using a retractable lanyard on a rooftop or container, the device should connect to the technician and not to the structure. This will avoid any reduction in the stopping distance. However, not all retractable lanyards are rated for this work.
- The technician has to be aware of the limitations of the system and keep a safe distance from edges where the device will not protect them.
- It is important to maintain a low body position.



5. **Fall Arrest**

A system used to suspend a person safely **after** a fall has occurred, in such a way as to prevent injury (as far as possible).

Fall arrest does **not** prevent a fall like work restraint does. It only prevents a person from serious injury or hitting the ground due to a fall by reducing the shock load.



i. Temporary vertical lifelines

Vertical lifelines are used for easy and quick movement up and down a structure. These lifelines are used along with a rope grab device which results in a reduced falling distance, the falling distances being limited only to the amount of rope stretch.

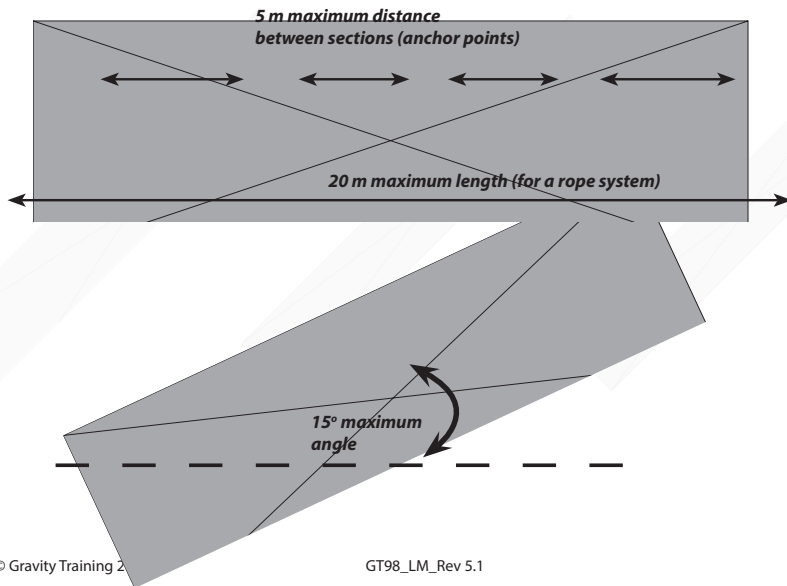
- Maximum length: 30 m.
- Maximum persons allowed on the lifeline: one person.
- Maximum angle between the two loops of the double-loop figure 8: **90°**.

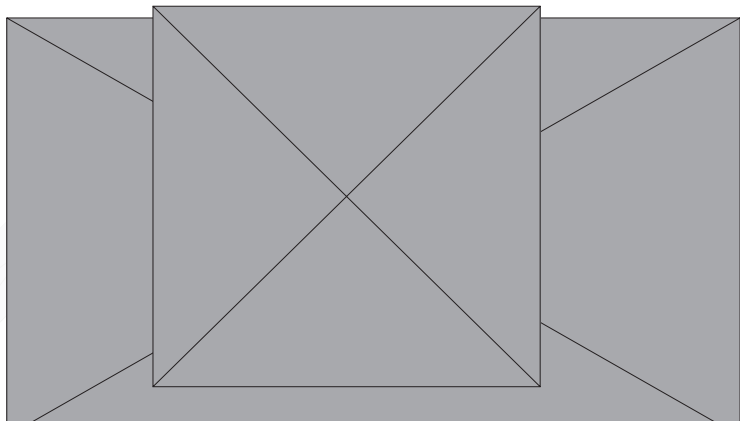


Temporary Vertical Lifeline

ii. Temporary horizontal lifelines

Horizontal lifelines are used for safe and quick movement from side to side. These lifelines should be used along with a shock absorber which can slide horizontally as the technician moves.



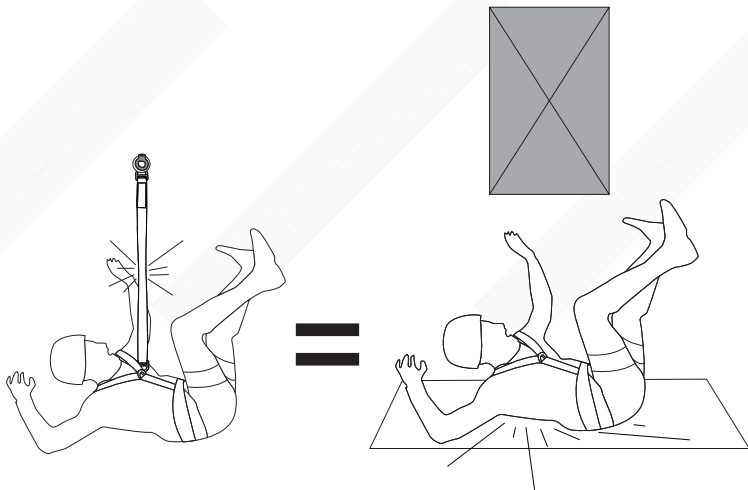


***No more than two users on a line .
No more than one user per section.***

C. WORK-AT-HEIGHT PRINCIPLES

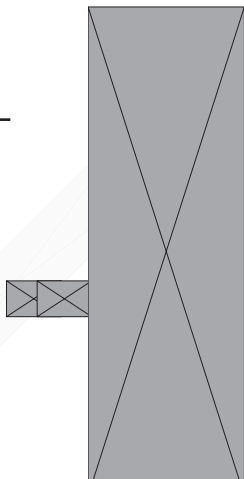
1. Shock Load

Shock load is the force imparted when a fall is arrested. High shock loads can lead to injury or death and must, therefore, be reduced or avoided.

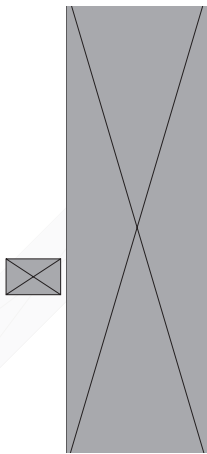


2. **Fall Factors**

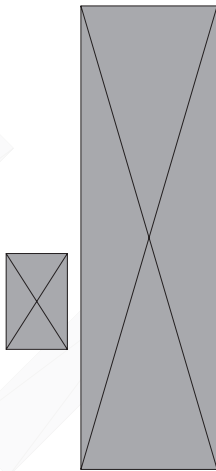
$$\text{Fall Factor} = \frac{\text{Distance of Fall}}{\text{Length of Lanyard}}$$



Fall Factor 0



Fall Factor 1



Fall Factor 2

3. Minimum Free Space

Minimum free space (MFS) is the open space a technician would need below them to safely arrest their fall without hitting any obstacles or reaching the ground. Reducing the falling distance or Fall Factor will reduce the required MFS.

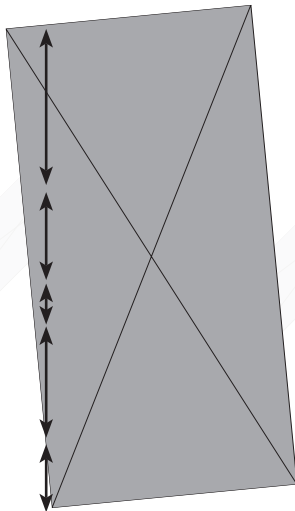
2 m for the lanyard length.

1.5 m for the shock absorber deployment. At Fall Factor 2 the lanyard will deploy completely.

0.5 m for the harness stretch.

1.5 m for the person's body.

0.5 m free space so that the person does not hit the ground.



$$\text{THUS: } 2 \text{ m} + 1.5 \text{ m} + 0.5 \text{ m} + 1.5 \text{ m} + 0.5 \text{ m} = 6 \text{ m}$$

4. Structural Evaluations

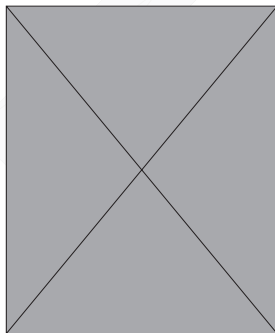
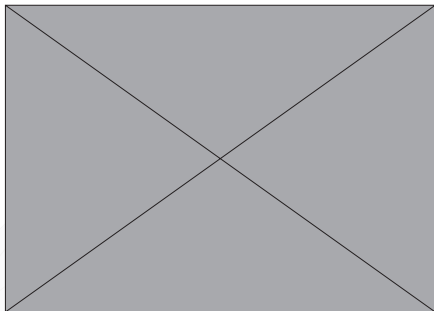
It is the responsibility of the owner of the structure to ensure that the structure is safe to climb when any rigging and lifting activities need to be performed.

All structures must be visually inspected before any climbing or lifting operations are conducted. It is not the responsibility of the technician(s) to certify the structure. However, the technician(s) must ensure that the structure is safe to climb.

Technicians must ensure that the system is SAFE:

- **Structure** is singed off by an engineer.
- **Access** methods are safe and secure. The surface areas of any roof structures have been inspected.
- **Fixed** fall arrest systems are certified and safe to use.
- **Ensure** that there are no missing members or loose telecoms equipment.

Note that a **permanent** fall arrest system must be installed by a competent person and be inspected once a year.

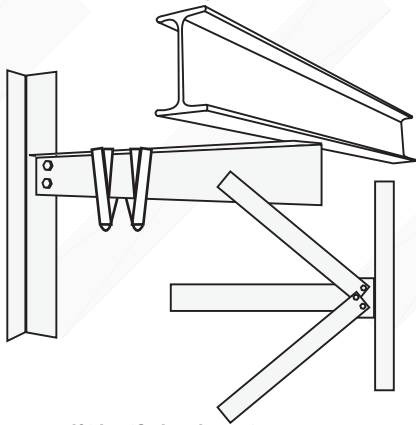


D. ANCHOR POINTS

Anchor points are intended to be the last remaining connection point between a worker and a structure. Should a fall occur, anchor points will prevent the worker from falling to the ground.

Anchor points are mainly divided into two categories:

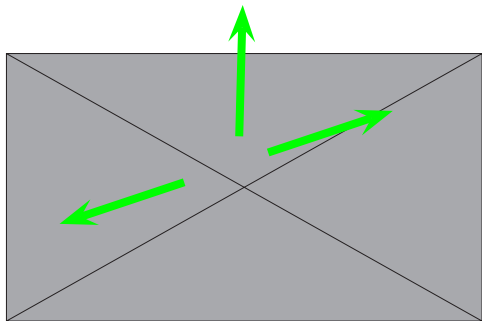
- Self-identified anchor points
- Planned anchor points (EN 795:2012) [To be used as per manufacturer's recommendations.]



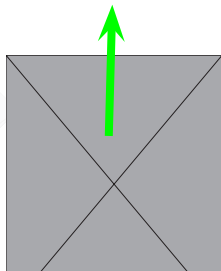
Self-identified anchor points:

SWL = 150 kg

MBS = 1 500 kg



Gravity Access Anchor



Eye-nut

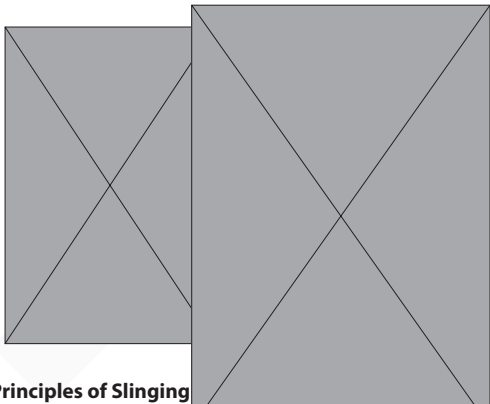
Planned anchor points:

SWL = 120 kg

MBS = 1 200 kg

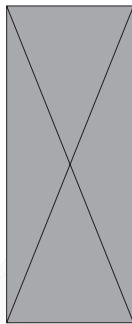
E. DIRECT LIFTING

Light materials weighing 8 kg or less should be connected to the harness gear loops. Light materials weighing in excess of 8 kg should be lifted using a separate 1:1 system with a maximum lifting capacity of 20 kg. Any light materials or equipment weighing more than 20 kg need to be lifted by a qualified rope rigger.



Principles of Slings

- Never exceed the SWL of equipment.
- Always protect slings from sharp edges.
- Always perform a pre-lift before lifting.
- Always consider the centre of gravity.



Principles of Rigging:

- Never stand underneath a load.
- Always ensure that the route traveled is clear of obstructions.
- Always ensure that the correct equipment is used for the lift.

F. EMERGENCY PROCEDURES

Ensure that a rescue plan is in place so that a rescue can be performed as quickly and efficiently as possible.

Perform a rescue when:

- a person has fallen and his shock absorbing lanyard has deployed,
- a person has lost consciousness,
- a person is unable to climb down to safety unassisted.

All rescues are different but there are a few things that stay the same throughout:

- Always call the emergency services **before** attempting a rescue.
- Always establish and maintain communication with the casualty throughout the rescue to determine their status, monitor their condition and prevent their condition from worsening. Two-way radios or hand signals can be used if the person is not within earshot.
- It is recommended that all rescuers are trained first aiders as well. First aid helps to prevent the casualty's condition from worsening.

Ratchet rescue procedure:

1. Call the emergency medical services.
2. Climb up to the casualty.
3. Connect the rope to the casualty's top D-ring.
4. Connect the suspension trauma sling.
5. Climb above the casualty.
6. Anchor the rescue ratchet with an attachment sling.
7. Remove all the slack between the casualty and the descending device/descender.
8. Lift the casualty with the rescue ratchet.
9. Remove the casualty's lanyard.
10. Apply extra friction.
11. Lower the casualty.

Self-escape rescue requirements (for towers/monopoles):

For this rescue, a self-escape rescue kit (to facilitate self-evacuation during an emergency) must be taken up by the first person who ascends the structure. The kit must be stored in a safe place (e.g. the crow's nest of a monopole).

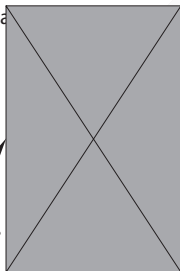
1. Ratchet Rescue

The ratchet rescue is not the only fall arrest rescue technique, but it is the one that will be focused on in this course. A double-rope rescue system should not be used because it is beyond the scope of fall arrest workers and endangers both the rescuer and casualty.

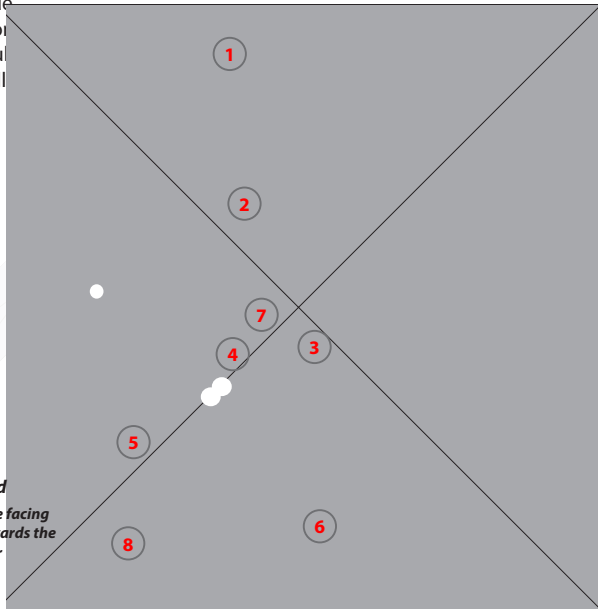
Rescue Kit:

1. Attachment sling
2. Rescue ratchet
3. Descending device
4. Kernmantle rope
5. Suspension trauma sling
6. Attachment karabiner
7. Extra-friction knot
8. Kit bag

Unlocked
Handle facing
down towards the
load



Locked
Handle facing
up towards the
anchor

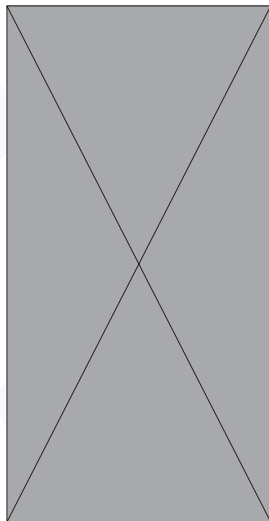


2. *Suspension Trauma*

The two biggest risks associated with awaiting a rescue while suspended in a harness are injury and death due to suspension trauma.

The main cause of suspension trauma is hanging stationary in a harness for a prolonged period. Due to the force of gravity, the blood in the fallen person's body is pulled towards the ground and pools in the legs. Due to the constriction of the leg loops of the harness, the blood cannot return to the fallen person's heart. This prevents blood from being circulated through a person's body. Thus, less and less blood reaches the brain, leading to loss of consciousness, which can lead to brain damage. Finally, this can lead to cardiac arrest and eventually death.

All of this can happen within 15–20 minutes; so, it is important to have a good knowledge of the techniques needed to prevent suspension trauma (for yourself and a casualty that needs to be rescued).



3. *Preventing Suspension Trauma*

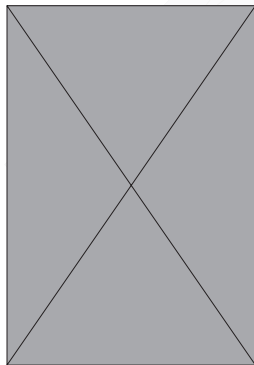
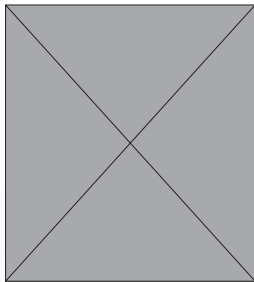
Muscle Pump:

Moving your legs, feet and toes will contract the muscles in the legs and force the blood back into circulation. This will provide the brain with the much-needed oxygen to stay conscious.

Suspension Trauma Sling:

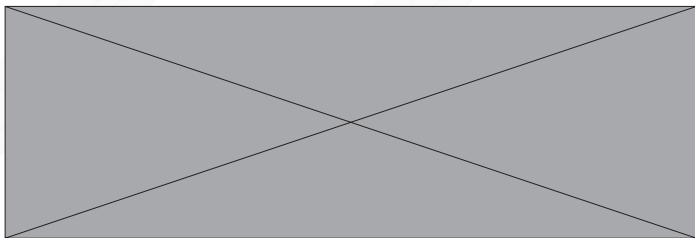
The sling will help you to stand up or to get into a horizontal position. This will prevent blood pooling in your legs and relieve the pressure of the leg loops, which will allow the blood to keep circulating through your body.

A rescuer may need to help an unconscious casualty to get into a horizontal position.



4. ***Supine Recovery Position***

Also known as the “Supine Position”. This position allows the blood to flow slowly enough to prevent Reflow Syndrome (a potentially fatal condition that can occur after blood pools in a part of the body for a prolonged period such as during suspension trauma). At the same time, it keeps the casualty stable and still to prevent further injury to the neck or spine.



G. FALL PROTECTION PLAN

Before any work at height may commence, all the fall risks onsite must be addressed and documented in an FPP.

The purpose of an FPP is to ensure the safety of workers.

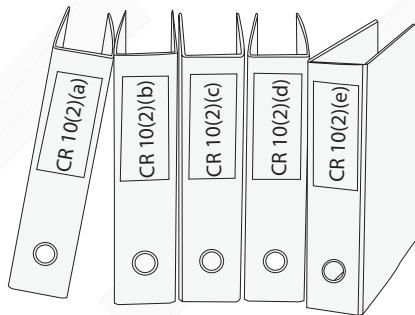
According to CR10(2), an FPP must include:

- CR10(2)(a) – **Risk Assessment**
- CR10(2)(b) – **Medical Fitness**
- CR10(2)(c) – **Training Programme**
- CR10(2)(d) – **Equipment Management**
- CR10(2)(e) – **Rescue Plan**

An FPP must also be reviewed if an incident occurs, if a change occurs onsite or in the case of a work scope change.

An FPP must be updated to ensure that all fall risks are mitigated and controlled onsite.

Note: All this must be done **before** work may start.



H. RISK ASSESSMENT

The most important part of an FPP is the **Risk Assessment**.

What to do when approaching a worksite:

STEP 1: RISK IDENTIFICATION

The following are factors that must be considered when identifying risks:

- **Site requirements**
- **Access method requirements**
- **Required team composition**
- **Fall protection requirements**
- **PPE requirements**
- **Environmental requirements**



After identifying the risks, the risks must be communicated to everyone onsite by means of:

- Site induction(s)
- Toolbox talk(s)

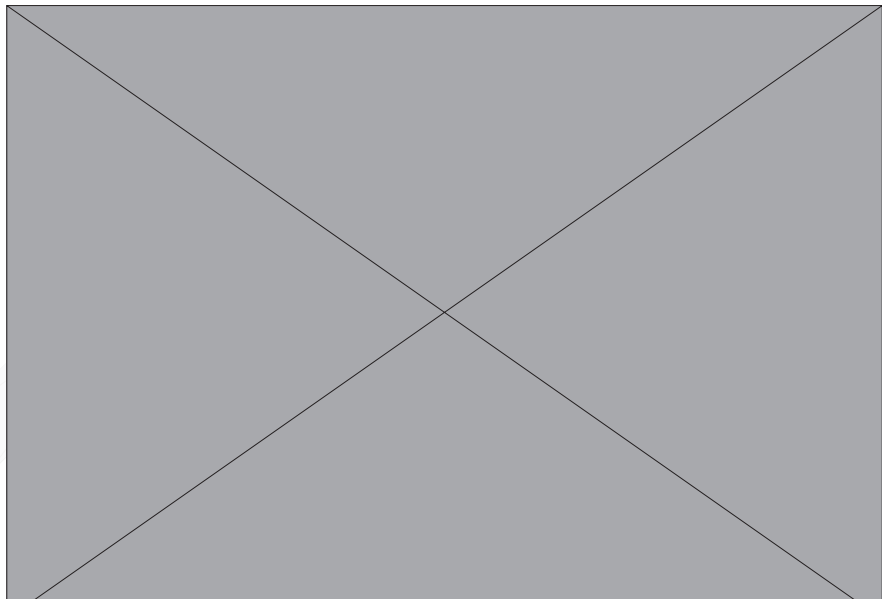
Remember that falling from height is the main possible hazard to avoid while working at height.

STEP 2: PRE-MITIGATION RISK RATING

Risk Matrix			Consequences or Severity				
			1	2	3	4	5
			<i>Negligible</i>	<i>Minor</i>	<i>Moderate</i>	<i>Major</i>	<i>Catastrophic</i>
Likelihood of Occurrence	1	<i>Rare</i>	LOW 1	LOW 2	LOW 3	LOW 4	MEDIUM 5*
	2	<i>Unlikely</i>	LOW 2	LOW 4	MEDIUM 6	MEDIUM 8	MEDIUM 10
	3	<i>Possible (50/50)</i>	LOW 3	MEDIUM 6	MEDIUM 9	MEDIUM 12	HIGH 15
	4	<i>Likely</i>	LOW 4	MEDIUM 8	MEDIUM 12	HIGH 16	HIGH 20
	5	<i>Almost Certain</i>	MEDIUM 5*	MEDIUM 10	HIGH 15	HIGH 20	HIGH 25

No work is allowed to commence if there is a **Medium** or **High** risk present onsite. Thus, we have to mitigate all identified risks in order for them to receive a **Low** risk rating.

STEP 3: RISK-MITIGATION HIERARCHY



STEP 4: POST-MITIGATION RISK RATING

Risk Matrix			Consequences or Severity				
			1	2	3	4	5
			<i>Negligible</i>	<i>Minor</i>	<i>Moderate</i>	<i>Major</i>	<i>Catastrophic</i>
Likelihood of Occurrence	1	<i>Rare</i>	LOW 1	LOW 2	LOW 3	LOW 4	MEDIUM 5*
	2	<i>Unlikely</i>	LOW 2	LOW 4	MEDIUM 6	MEDIUM 8	MEDIUM 10
	3	<i>Possible (50/50)</i>	LOW 3	MEDIUM 6	MEDIUM 9	MEDIUM 12	HIGH 15
	4	<i>Likely</i>	LOW 4	MEDIUM 8	MEDIUM 12	HIGH 16	HIGH 20
	5	<i>Almost Certain</i>	MEDIUM 5*	MEDIUM 10	HIGH 15	HIGH 20	HIGH 25

No work is allowed to commence if there is a **Medium** or **High** risk present onsite. Thus, we have to mitigate all identified risks in order for them to receive a **Low** risk rating.

I. HAZARDOUS ENVIRONMENTAL CONDITIONS

1. General Information

No work at height shall be conducted during any inclement weather conditions. Should the weather condition change whilst at height, all technicians must safely descend to the ground and the site supervisor must report back immediately to the project manager.

Examples of inclement weather conditions:

- Rain
- Lightning
- Extremely cold weather
- Extremely hot weather
- Sleet or snow
- High or gusty winds
- Poor visibility (e.g. fog or smog)

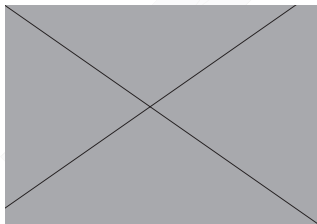
Examples of other hazardous conditions:

- Fumes
- Smoke



2. **Bee Awareness**

- Honeybees are an endangered species that is often a hazard when working at height. They are responsible for pollinating 70 of the 100 crop species that feed 90% of the world and, as such, are a protected species that may not be killed.
- When starting work on a new site, always look out for bees in the area where work will take place. Multiple bees usually fly around the area where a nest is located, and this serves as a warning for intruders. Bees are most active during the day, making the chances of an attack much higher. They are more docile at night and in colder temperatures but will still attack if disturbed.



Dos	Don'ts
<i>Stop work if it is suspected that there are bees on the site.</i>	Purchase a bee suit so that the work can continue.
<i>Observe the area where the bees are suspected and keep a safe distance (30 m+ from the swarm).</i>	Approach fast or in an aggressive manner, or make sudden movements.
<i>Take pictures from a distance to assist the beekeepers in finding where the swarm is located.</i>	Burn or smoke the bees out or poison them in an attempt to kill the swarm.
<i>Listen for and adhere to warnings from the bees. (Bees will typically buzz frantically and bump into people to warn them to stay away.)</i>	Try and remove the swarm if you are not a trained and experienced beekeeper.
<i>Stay calm in the event of an attack and walk away briskly.</i>	Swat the bees, throw objects at the bees or run around screaming.
<i>In the event of an attack, keep your head and face covered (especially your mouth).</i>	Jump into water.
<i>Take cover (e.g. inside a vehicle or building) when you have reached a safe distance.</i>	Return to the area where the attack occurred before the swarm has been removed.
<i>Use a scraper or flat object to remove stingers.</i>	Squeeze or pinch to pull out stingers.
<i>Seek emergency treatment after an attack, especially if there is no EpiPen available or if the casualty is showing symptoms of anaphylactic shock.</i>	Leave the stinger(s) attached, and don't brush off or ignore symptoms of anaphylactic shock.

Removing bees is a specialized task as it can be fatal if done by incompetent and/or inexperienced technicians. Bee removal should only be done by designated, competent and authorized beekeepers.

3. **Bird Awareness**

Crows

- Crows are found on every continent except Antarctica, and are among the most intelligent animals on the planet.
- They are able to recognize a single person's face among many other people.
- They are able to communicate to other crows and share information.
- Crows normally attack humans because humans enter the birds' habitat. During breeding season, crows defend their eggs or young ones against humans that threaten them.
- If you see a nest or you see a crow on the structure, do not advance. They usually make a noise to warn you before they attack.
- Contact a professional to take care of the crow(s) before work commences.



Hawks and Eagles

- Hawks and eagles are also among the most intelligent birds on the planet, and they are also a danger to people working at height.
- The most common reason for an attack is that a human comes too close to a bird's nest during breeding season, posing a danger to the bird's eggs or young ones.
- By entering their habitat, you automatically pose a threat to them because they think you want to injure them or take their food and/or water. Since it is within their nature to protect themselves, they are likely to attack if you enter their habitat.
- Hawks and eagles will usually warn you by making noise before they attack.
- Do not attempt to deal with them by yourself. Get a professional to take care of the bird(s).
- Inform your manager about the presence of the birds and take photos as proof.



J. SPECIALISED CONTENT

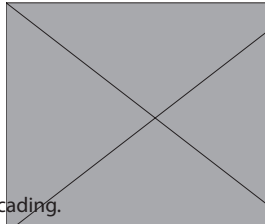
1. Roof Work

Roof work entails the deed of installing, removing, repairing or otherwise maintaining any covering above ground level whether on a roof or in a roof (e.g. building a roof with tiles or zinc plates, waterproofing, painting, installing and maintaining solar systems, plumbing or gutter/chimney cleaning).

Working on a roof is a dangerous activity and many fatalities occur due to workers not being aware of the risks and not mitigating the relevant risks properly.

Main hazards associated with roof work:

- Falling from a flat or sloping roof due to inadequate edge protection or barricading.
- Falling through fragile/damaged roof surfaces (e.g. skylights or ceilings).
- Objects falling from a roof due to inadequate edge protection or objects not being tied off.
- Falling from a ladder or alternative access method when accessing a roof.
- Manual handling of materials (where factors like size, weight, temperature and chemical nature present a challenge).

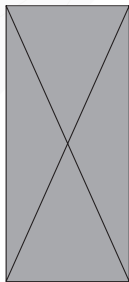


Every roof worksite will have challenges. The material of the roof, the access method to the roof and the safety system all play a huge role and will have to be assessed before work starts.

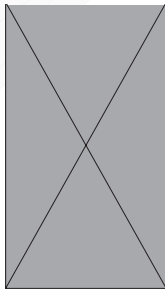
Should there be any question regarding the integrity of the structure holding up the roof (trusses, beams etc.) or the roof covering (iron sheeting, tiles etc.), work should be halted until a structural engineer has confirmed that the roof is safe to work on.

Portable Ladders

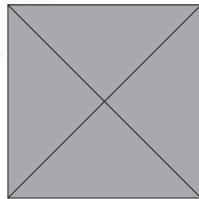
Portable ladders are the most common and inexpensive way to gain access to rooftops. However, it is also considered among the most dangerous methods of roof access if the worker is not properly trained. "Portable Ladder User" training is recommended.

**Scaffolding**

Scaffolding is one of the most expensive and time-consuming roof work options. Scaffolding must be assembled by a competent erector and signed off by a competent inspector.

**Mobile Elevated Work Platforms (MEWPs)**

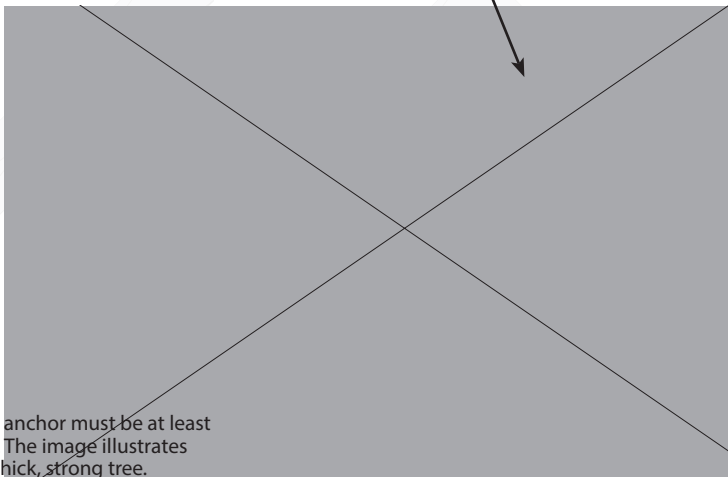
MEWPs are probably the safest method for roof work. However, it is an expensive method and the site does not always allow for a MEWP to gain access or to move around. A competent MEWP operator needs to operate the MEWP at all times.



ii. *Safety systems while working on roofs with anchorage methods*

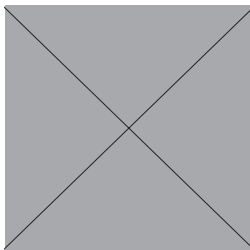
NB: The anchorage methods used in the system below have to meet the requirements of EN 795 type B or E. The system limits the user to only be on one side of the roof and not go within 0.5 m of the ridge (top of a sloping roof).

Please note that the following methods can only be used for a pitch of no more than 30 degrees. Any slopes steeper than 30 degrees require rope access techniques.



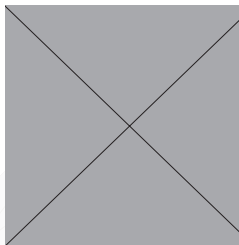
iii. *Various anchorage options*

Anchoring to a vehicle



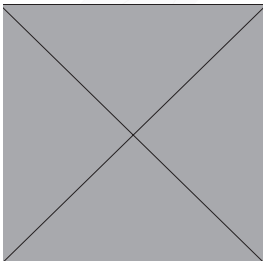
This system is installed on the vehicle axle. The climber should be in possession of the vehicle keys.

Anchoring to a weight



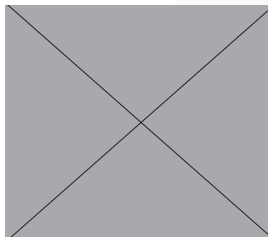
This system is installed on a weight designed to carry the load of a self-identified anchor point.

Anchoring to a Duckbill anchor



This system is installed into the ground and then pulled to lock the Duckbill anchor, which should have a wire sling attached.

Anchoring to two trusses of the roof

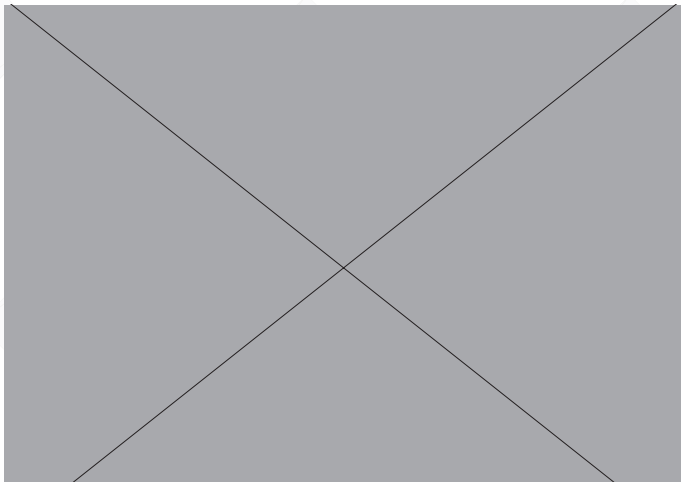


This system is installed on two of the trusses to share the weight between the anchor points.

iv. Low-level fall arrest system (FAS)

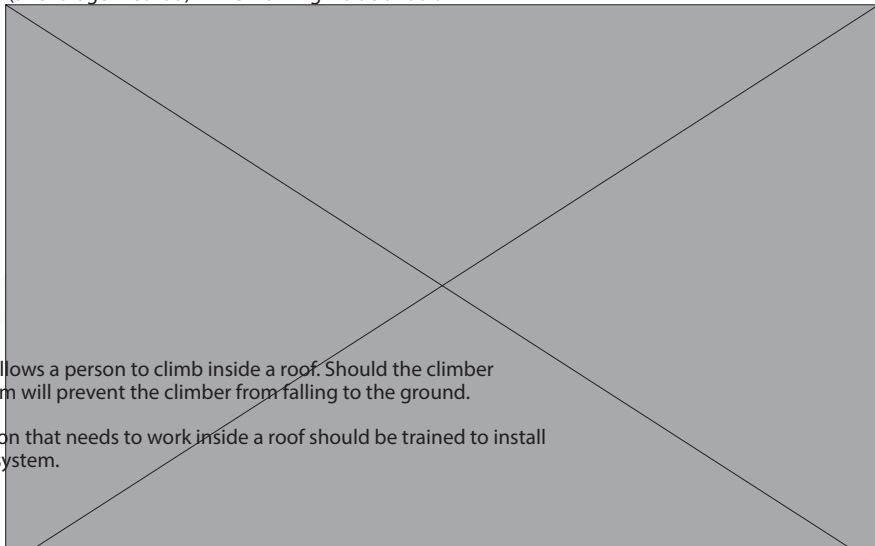
NB: The low-level FAS below has to meet the requirements of EN 795 type B or E. It allows the user to move to any side of the roof.

The user will connect themselves to the lifeline with the use of a rope grab device. The lifeline is connected to the knot in the middle of the system. The two descenders attached to the bottom anchors enable the tensioning of the system and allow a rescue to be performed at any stage.



v. *Working inside a roof*

Safety system (anchorage method) while working inside a roof:



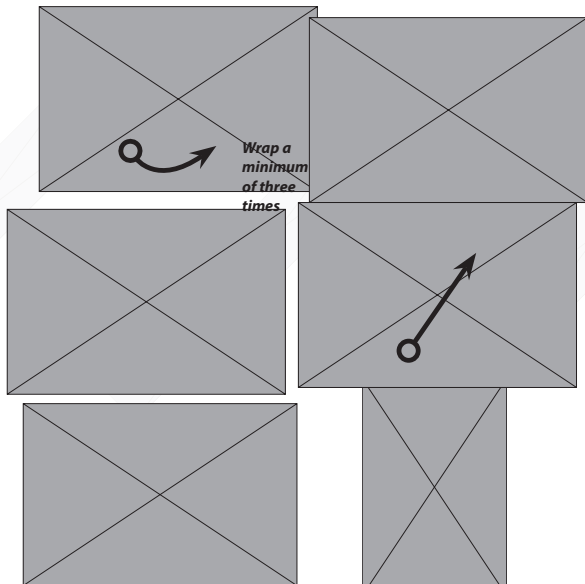
This system allows a person to climb inside a roof. Should the climber fall, this system will prevent the climber from falling to the ground.

NB: Any person that needs to work inside a roof should be trained to install and use this system.

K. KNOTS

1. Klemheist

This knot is used to create an anchor point on a vertical or diagonal pole. It should not be used in a Fall Factor 2.



L. THE WAY FORWARD

Congratulations on completing Gravity Training's Fall Arrest and Basic Rescue course.

If found not yet competent (NYC), you will need more training before you can be reassessed. Please do not be discouraged. Not everyone passes on their first attempt. Some learners need a bit more experience to obtain the required knowledge and skills. You will receive comprehensive feedback on your performance and be informed on how to proceed.

At Gravity, we care about the professional development of our clients. Should you wish to take the next step, consider enrolling for some of our other courses. See The Way Forward and the Gravity Skills and Development Map or consult your facilitator/assessor.



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